

Muse

Structured Field Intelligence from Noisy Bangla Speech and Handwriting

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A voice- and image-driven capture pipeline that recovers reliable structured data from Bangla field reports at word error rates near 34%, using a deterministic linguistic layer and constrained generation rather than a larger acoustic model.

Abstract

Bangladesh’s FMCG sector is a roughly USD 4 billion market in which about 97% of trade flows through traditional retail: small, independent shops visited in person by distribution representatives. Sales-force automation captures the transaction from those visits but not the observation—why a shop refused an order, what a competitor is running, what a retailer complained about. That information is never recorded, because typing several sentences at forty outlets a day is not feasible.

Muse captures it by voice and by photograph of the handwritten order note. The central difficulty is that Bangla speech recognition, which reports 3–5% word error rate on clean read benchmarks, degrades to approximately 34% on real long-form field audio. Rather than pursue a better acoustic model, Muse treats the transcript as unreliable by design and recovers the *fields* through a deterministic layer: a Bangla quantity grammar covering the fractional and multiplicative forms that general models mishandle, and phonetic normalisation that folds Bangla orthography and Latin catalogue names into a shared key space. A language model is then constrained per recording to a vocabulary the resolvers actually produced, making an unresolvable product inexpressible rather than merely discouraged. Confidence is derived from upstream evidence, never self-reported, and gates a one-tap clarification returned to the representative. The system comprises 412 automated tests; the deterministic layer accounts for roughly 40 ms of a 2.4 s pipeline.

1 Problem

A distribution representative in Dhaka observes a competitor promotion running across twelve shops on a Tuesday. Under current practice his brand manager learns of it weeks later, inferred from a sales decline. The delay is not a failure of diligence; it is structural. Sales-force automation systems record order quantity, SKU, outlet and timestamp. They provide no mechanism for the unstructured observation, and no realistic mechanism could: the representative has fifteen seconds per shop and both hands occupied.

The consequence is measurable. Published estimates place the global average retail out-of-stock rate at approximately 8.3%, and find that 58% of shoppers encountering a stock-out do not defer the purchase but abandon it. Shortening the interval between a field observation and a decision is therefore a revenue protection problem, not a labour cost problem—a distinction that matters in a market where a representative’s monthly wage does not justify automation on cost grounds alone.

1.1 Why existing approaches do not address it

Approach	Captures	Does not capture
Sales-force automation	Orders, GPS check-in, route compliance	Any unstructured observation
Retail audit panels	Rigorous, comparable measurement	Sample-based and monthly; reports the previous period
Messaging groups	Ad-hoc photographs and text	Unstructured, unsearchable, not aggregable
Additional headcount	More visits	The same capture limitation, multiplied
General speech-to-text	A transcript	Product resolution, confidence, Bangla numerals

Table 1: A transcript is not data. The gap is between recognised text and a resolved, checkable field.

2 System architecture

Muse is a six-stage pipeline. Two stages call external models; the remaining four are domain logic. Both capture modalities—voice and photograph—diverge for exactly one stage and share everything after it.

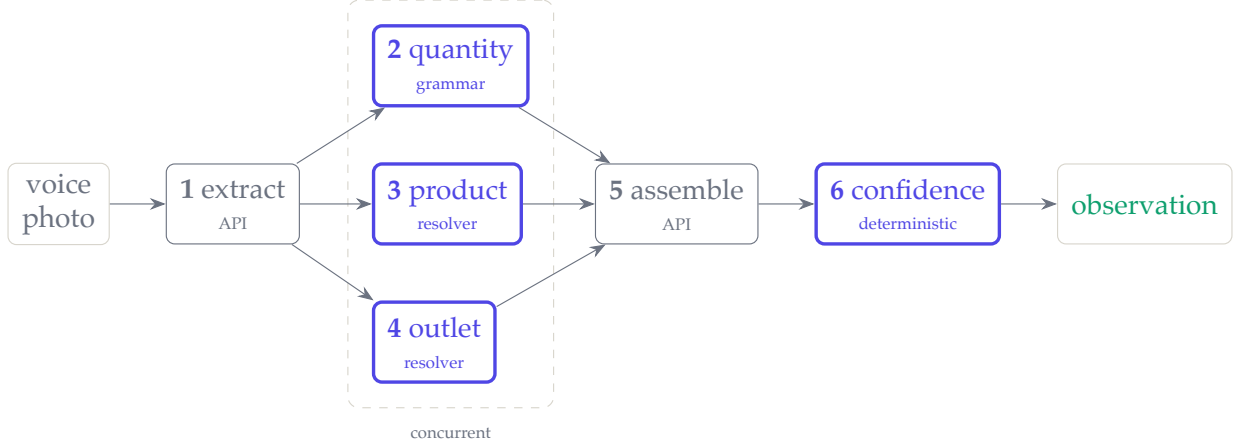


Figure 1: Stages 2–4 are independent and run concurrently; they annotate a shared transcript without modifying it. Blue stages are domain logic; grey stages are external model calls.

Two invariants hold throughout. First, stages annotate but never rewrite the transcript: each annotation carries character offsets into the original text, so a later stage can inspect the evidence that produced a value. Second, the extraction stage is the only point at which voice and photograph differ, which is what allowed a second modality to be added by modifying four files.

3 Technical approach

3.1 A Bangla quantity grammar

Bangla expresses quantity with fractional prefixes and multiplicative units that general-purpose models handle unreliably. The grammar resolves these deterministically.

Form	Fractional		Form	Multiplicative	
	Value		Value		
দেড়	1.5	standalone	ডজন	$\times 12$	dozen
আড়াই	2.5	standalone	হালি	$\times 4$	set of four
সাড়ে X	$X + 0.5$	prefix	কুড়ি	$\times 20$	score
সোয়া X	$X + 0.25$	prefix	গুণ্ডা	$\times 4$	
পৌনে X	$X - 0.25$	subtracts	লাখ	$\times 10^5$	

Table 2: পৌনে subtracts, which naive keyword matching inverts. দেড় ডজন evaluates to $1.5 \times 12 = 18$ pieces.

3.2 Phonetic normalisation

Bangla orthography maps many graphemes onto one sound, and speech recognition loses these distinctions first. Both the recognised text and the product catalogue are folded into a shared phonetic key space, which also absorbs the script boundary: catalogues are written in English while representatives speak Bangla.

Two comparison modes are required. Product and outlet names use a lenient comparison that also considers the consonant skeleton, since vowels are the least reliable signal. The numeral

Collapse	Key	Heard	Resolves to
শষস	s	হইল → huil	Wheel → huil
ণন	n	প্রান → pran	PRAN → pran
রঙঢ়	r	ম্যাঙ্গো → mango	Mango → mango
vowel length, aspiration	—	তীন → tin	তিন → tin

Table 3: Word-initial Latin “wh” maps to hu, since Bengali renders it হু. Without this the brand never matches its own catalogue entry.

lexicon uses a strict comparison retaining vowels: সতেরো (seventeen) and স্টোরে (“at the store”) share the skeleton s-t-r exactly, and a lenient comparison reads a shop as a number.

3.3 Constrained generation

A language model performs segmentation and semantics only. Its response schema is rebuilt for every recording, with identity fields restricted to an enumeration of exactly the candidates the resolvers produced:

```
skuId: enum(["SKU-404", "SKU-407"]).nullable()
```

A product that was not resolved is therefore not expressible. This is a structural constraint rather than an instruction, and it is what makes the system deployable where a confidently incorrect SKU is worse than no record. Numeric fields receive a second guard, since a schema cannot bind a float: any quantity the model returns is checked against what the grammar parsed and discarded otherwise.

3.4 Derived confidence and clarification

Confidence is computed from upstream evidence and never requested from the model, which is poorly calibrated and will assign high confidence to invented values.

$$\text{conf}(f) = \text{asr}(\text{span}_f) \times m(\text{margin}) \times g(\text{grammar})$$

The margin term—the gap between the resolver’s first and second candidate—is decisive. A high top score with a narrow margin indicates that two products are indistinguishable, and gating on score alone would confirm precisely those cases.



Figure 2: Low confidence sets a status; it never discards a record. A dropped observation is unusable in an enterprise, whereas a flagged one is merely honest.

An unanswered prompt resolves to the best guess after a fixed interval, and an answer arriving after that resolution still applies—the record was already published with a guess, so a late correction must reach the dashboard.

4 Implementation and results

The system is implemented in TypeScript across an Express API, a queue worker and a React client, with MongoDB and Redis. Pipeline stages are dependency-free classes constructed with ports, so the test suite requires no container.

4.1 Worked example

The following is verbatim output from a production speech model on real audio.

Recognised: বজোই স্টোর যে প্রান মাল্পো জুস দেব দর্জন লগেগা

Recovered: Bijoy Store (0.79) · PRAN Mango Juice 250ml (0.98) · 18 piece (0.85)

Four words are mis-transcribed—the outlet name, the brand, and the numeral phrase—and every field is nonetheless correct. This is the intended behaviour rather than a fortunate case: the resolvers never consult the transcript for correctness, only for similarity against a closed catalogue.

4.2 Measured characteristics

Measure	Value	Note
Automated tests	412	full suite under 1 s, no network
Deterministic layer latency	≈40 ms	of a ≈2.4 s pipeline
Extraction latency	450–1250 ms	external model call
Assembly latency	≈1.9 s	external model call
Files changed to add photograph capture	4	

Table 4: Nearly all latency is external model calls; the domain logic is negligible by comparison.

5 Economic model

The following is a model, not a measurement, and is presented with its assumptions stated. For a brand covering 10,000 outlets at BDT 25,000 monthly sell-through each (BDT 3.0 billion annually):

Out-of-stock rate	8.3%	published
Share becoming lost sales	58%	published
Revenue lost to stock-outs	4.8%	derived
	BDT 144 M	per year
Reduction from same-day reporting	20%	assumed
Recovered	BDT 28.8 M	per year
Platform cost (150 representatives)	BDT 2.16 M	per year
Return	≈13×	

Table 5: Only the 20% reduction is assumed; halving it still yields roughly a sixfold return.

6 Limitations and further work

- **Dialect.** Version one targets Dhaka-standard Bangla. Chittagonian and Sylheti are effectively distinct languages for a speech model and are explicitly out of scope.
- **Handwriting recognition is simulated** in the current build. Text extracted from a photograph is canned; every stage after extraction is the production pipeline, and this is disclosed in the interface itself.
- **Evaluation.** A harness measuring word error rate, per-field precision and confidence calibration is implemented and gated on regression, but requires a labelled corpus of field recordings that has not yet been collected. No accuracy figure is claimed here for that reason.

- **Unmatched surface forms** produce no annotation and therefore cannot enter the alias-learning queue—precisely the case where an alias would help most. Addressing this requires the resolver to emit sub-threshold near-misses.

Planned work includes domain-adapted Bangla speech recognition through n-gram shallow fusion over the customer catalogue, a learned confidence model trained on the evaluation corpus, and genuine optical character recognition for printed signage. The architecture generalises to other Bangla-speaking field forces—pharmaceutical representatives, microfinance officers, agricultural extension workers—by substituting the catalogue and observation schema.

References

1. The Business Standard / FICCI. Bangladesh FMCG market size and composition.
2. IHL Group. *Retail Inventory Crisis Persists Despite \$172 Billion in Improvements*, 2025.
3. Zebra Technologies. *Global Shopper Study*—stock-out conversion to lost sales.
4. FLEURS and Common Voice Bengali benchmarks; published long-form Bangla ASR evaluations.